
Design, Simulation, and Analysis of High-Quality Factor Fabry-Perot Cavity Resonators using SiEPIC-EBeam PDK and Lumerical Software Suite

Aryan Chahardovalee
Department of Electrical and Computer Engineering
University of British Columbia

1 Introduction

In this project, a high-quality optical resonator with a quality-factor above 100,000 is designed using the SiEPIC - EBeam Process Design Kit (PDK). Optical resonators are one of the three main components of a laser (resonator, source, and gain medium), and a high-quality resonator can significantly improve lasing performance.

The approach taken in this project involves designing a Fabry-Perot Cavity resonator by placing a waveguide between two equally-lengthed Bragg gratings. The governing equations for the quality factor of an optical resonator are presented below:

$$Q = \left(\frac{2\pi c}{\lambda_0} \right) \tau_p = \frac{\lambda_0}{\Delta\lambda_{3dB}}, \quad \tau_p = \frac{n_g}{c(\alpha - \alpha_m)}, \quad \alpha_m = -\frac{\ln R}{L_{cavity}}, \quad R = \tanh^2(\kappa N \Lambda) \quad (1)$$

In these equations, Q represents the quality factor, $\Delta\lambda_{3dB}$ denotes the 3dB bandwidth of the peak resonance, and λ_0 refers to the center frequency, which is 1310 nm in our design. The photon lifetime inside the cavity, τ_p , depends on the group index (n_g), waveguide propagation loss (α), and mirror losses (α_m). Minimizing mirror losses is the focus of our design in this project. Mirror loss is influenced by the reflectivity of the Bragg grating and is inversely proportional to the cavity length (L_{cavity}). A longer cavity length is expected to reduce mirror losses and consequently enhance the quality factor. Furthermore, when reflectivity reaches its maximum value, $\ln R$ yields 0, which again minimizes mirror loss. To maximize R , the argument inside $\tanh^2$ must approach infinity. This insight guides our project to maximize kappa, the grating number (N), and the grating period (Λ) as much as possible.

Enhancing the reflectivity of Bragg gratings, for instance, by increasing the grating length, generally leads to a higher Q -factor due to the increased amount of light reflected back from the resonator, which reduces the loss of light. As the Bragg gratings function as mirrors, they reflect light back into the waveguide, allowing it to bounce back and forth and be stored in the resonator for more extended periods. Consequently, bolstering the Bragg grating strength typically yields a higher Q -factor, which in turn can produce a narrower free spectral range and increased output power from the laser. Nonetheless, there is a threshold to the extent that Bragg grating strength can be amplified before other factors, such as scattering losses or nonlinear effects, start to dominate and impose limitations on the Q -factor. In this project, the Bragg number (Bragg length) and waveguide length are strategically varied to optimize resonator performance.

Lumerical MODE simulation software is employed to characterize the waveguide properties of this cavity, while Lumerical FDTD simulation software is used to simulate the Bragg gratings on either side. A Python model of the resonator, based on the waveguide transfer matrix, is developed, complemented by a Lumerical INTERCONNECT simulation of the entire optical circuit.

Table 1: Geometrical Parameters Summary

Name of Parameter	Value of Parameter	Type of Parameter
Width	Actual: 335 nm / fabricated: 350nm	Fixed parameter
Corrugation Width	30 nm	Fixed parameter
Thickness	220 nm	Fixed parameter
Bragg Number	Range from 100 to 200 – steps of 10	Varying parameter
Length of the Waveguide Cavity in between the Bragg Gratings	Varies from 100 μm to 40 μm – steps of 5.38 μm	Varying parameter
Bragg Period *	269 nm or 275 nm	Varying parameter

* Bragg period is alternated between 269 and 275 in separate designs to check for possible shifts in the central frequency.

The models will be fabricated and tested using Electron Beam (EBeam) Lithography techniques. Due to the fabrication process bias, the waveguides' actual width will be 335nm, 15nm smaller than the specified 350nm width. To account for this biasing effect, the Fabry-Perot Cavity in this project is designed for a width of 335nm and drawn for fabrication at 350nm.

2 Waveguide Design and Simulation

To obtain a reliable model of the waveguide characteristics that are to be used between the two Bragg gratings, we need to perform a simulation of a small slice of the waveguide and use the obtained values to build our Transfer Matrix Method (Python model) and Lumerical INTERCONNECT Compact Model for Bragg Grating in Section 4.

2.1 Waveguide Simulation

A waveguide width of 335 nm and thickness of 220 nm are chosen. This waveguide will support light with a 1310 nm wavelength. The cladding of the waveguide is Silicon Dioxide (SiO_2), and the core material is Silicon (Si). Simulations and frequency sweeps are done for the lowest TE mode. The results of the simulation are shown in Figure 1. The colour axis is in dB scale.

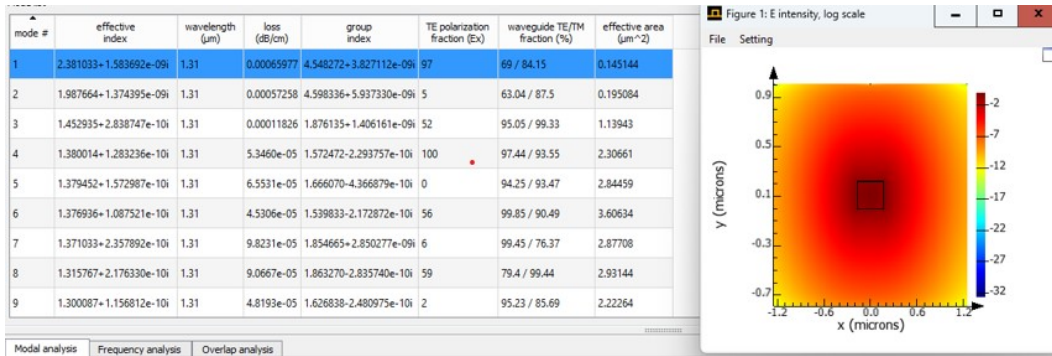


Figure 1: Lumerical MODE simulation for a 335 nm wide waveguide

It can be observed in the E-field intensity diagram in Figure 1 that the field has decayed sufficiently, and our results are reliable. For the lowest TE mode, the effective index (N_{eff}) is calculated to be 2.381033, and the group index (N_g) is 4.548272.

2.2 Neff vs. Wavelength Characteristics

A frequency sweep must be done on this model to obtain the graph for the changes in the effective index (Neff) vs. wavelength. The results are shown in Figure 2.

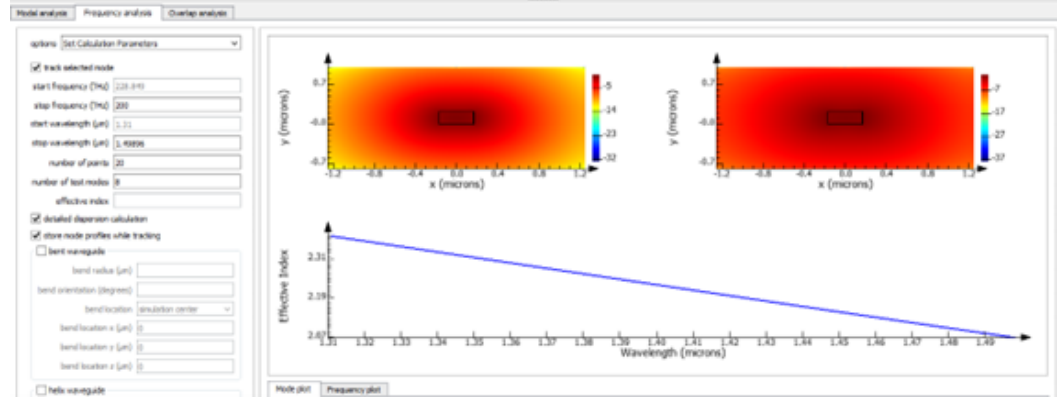


Figure 2: Frequency Sweep to Obtain Neff vs. Wavelength Characteristics

2.3 Simulation Results for the Standalone Waveguide

Upon performing a successful sweep, a script (Appendix: A1) provided by the instructor, will fit the sweep data and provide us with the coefficients of a parabolic function that will predict the behaviour of Neff vs. wavelength. These coefficients come from the graph in Figure 2 and are as follows:

$$a_1 = 2.381220, \quad a_2 = -1.669500, \quad a_3 = 0.113343$$

The equation which uses the above coefficients to predict the behavior of neff vs. wavelength is as follows:

$$\text{neff}(\lambda) = a_1 + a_2(\lambda - \lambda_0) + a_3(\lambda - \lambda_0)^2, \quad \lambda_0 = 1310 \text{ nm} \quad (2)$$

Table 2: Lumerical Mode simulation results

Parameter	Simulation Result
Group Index (Ng) – lowest TE mode	4.548272
Effective Index (Neff) – lowest TE mode	2.381033
Neff vs λ characteristic equation	$\text{neff}(\lambda) = 2.38 - 1.66 \cdot (\lambda - \lambda_0) + 0.11(\lambda - \lambda_0)^2$, $\lambda_0 = 1310 \text{ nm}$
Assumed Waveguide Loss	700dB/m

3 Bragg Grating Design and Simulation

An advanced Lumerical FDTD simulation is conducted on a single period of the Bragg grating. The Bragg grating features a width of 335 nm and a thickness of 220 nm, with a corrugation width fixed at 30 nm. The effective index and group index of the waveguide have been previously determined in Section 2. Utilizing formula (3), the Bragg grating period is computed to be 275 nm. It is crucial to note that if the simulations in Part 2 were executed using a 350 nm width (without considering the fabrication bias), the effective index would be 2.43, corresponding to a Bragg period of 269 nm. Consequently, the fabricated Fabry-Perot resonators in my design include multiple iterations with distinct Bragg periods to account for this discrepancy and better understand the effects of the fabrication bias process.

$$\lambda_B = 2 \cdot n_{\text{eff}} \cdot \Lambda \quad (3)$$

Through the simulation, we have determined the grating strength (κ) to be 114777, which is a crucial parameter for the performance of the Bragg grating. Additionally, the simulation results confirm that the Bragg grating's central frequency is in close proximity to the target 1310 nm, and the bandwidth is sufficiently broad to ensure optimal performance. These findings are illustrated in Figure 3 and Table 3. Furthermore, as depicted in Figure 4, the calculated propagation loss for the Bragg grating period under consideration is 0.63285 dB/cm. These results are consistent with the expected performance of Bragg gratings in similar configurations, demonstrating the efficacy of the chosen design parameters.

Table 3: Lumerical FDTD simulation results for the Bragg grating with 275 nm period

Parameter	Simulation Result
Grating Strength (κ)	114777
Bandwidth	14.2141 nm
Central Wavelength	1330.24 nm
Loss	63.285 dB/m

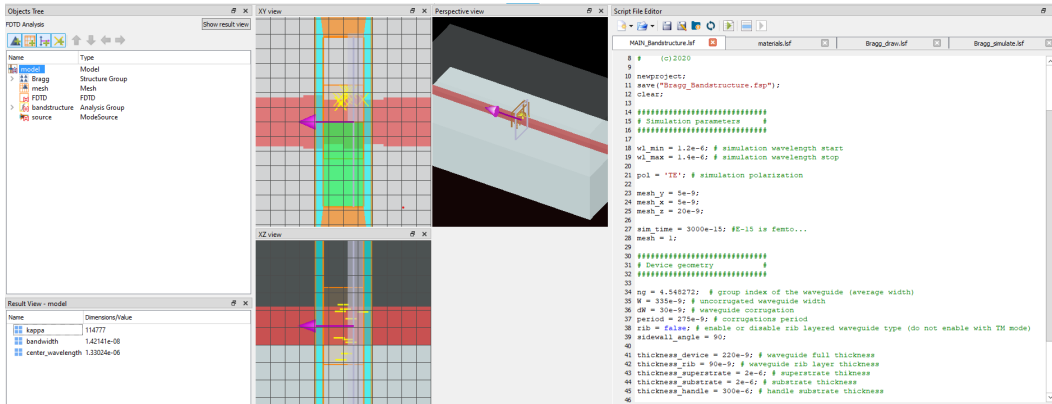


Figure 3: Lumerical FDTD simulation of the Bragg grating

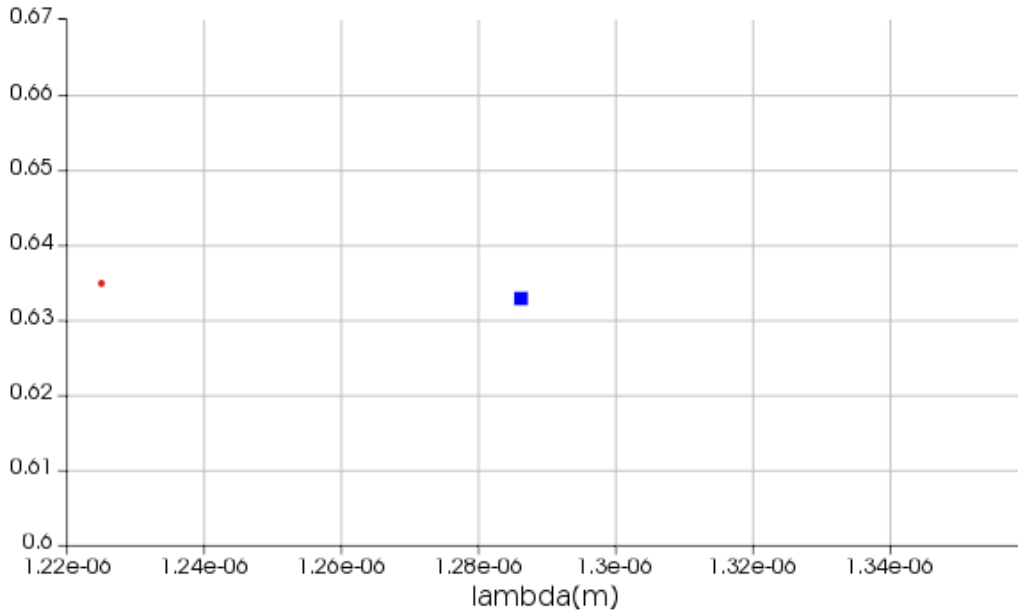


Figure 4: Lumerical FDTD calculated loss for the Bragg grating in dB/cm

4 Fabry-Perot Cavity Resonator Design and Simulation

To combine the Bragg grating design and the waveguide cavity, two different approaches are taken: A Transfer Matrix Method (using Python) and a Lumerical INTERCONNECT simulation.

4.1 Transfer Matrix Method

The Transfer Matrix Method (TMM) is a powerful numerical technique used to analyze and solve problems in linear optics and wave propagation, particularly in stratified media like multilayer structures and waveguides.

The fundamental idea behind the TMM is to represent the optical properties of each layer in a stratified medium by a matrix (called a transfer matrix). These matrices describe the relationships between the incoming and outgoing waves at the interfaces of the layers. By multiplying the transfer matrices of each layer, we obtain a single matrix representing the overall system. This resulting matrix then allows us to calculate the transmission and reflection coefficients of the entire structure.

According to equation (1), to obtain a high-quality factor, reflectivity should be at a maximum while the waveguide and mirror loss (in our case Bragg grating loss) have to be at a minimum. The Python model can predict the behaviour of the reflection and transmission as well as the central frequency accurately. Figure 5 depicts the predictions made by the TMM script for a 100-micron cavity with two Bragg mirrors on each side. Bragg gratings that are used to obtain Figure 5 have a grating number of 110 and a Bragg period of 275 nm (Design ARYANC3_1).

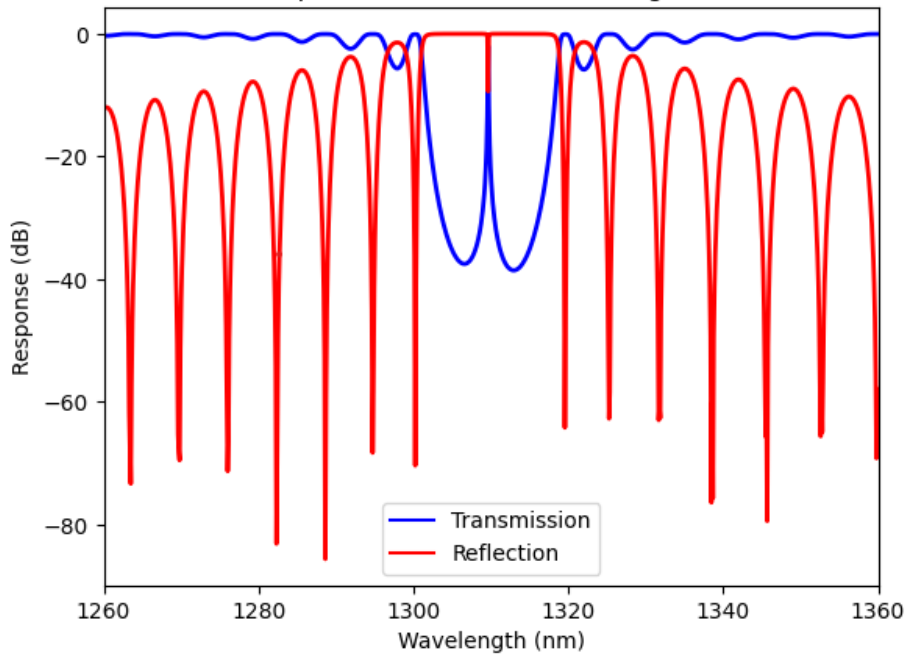


Figure 5: Calculated response of the structure using TMM (dB scale)

4.2 Lumerical INTERCONNECT

A Lumerical INTERCONNECT simulation employing simple library models, including two unidirectional Bragg gratings, an analyzer, and a basic waveguide, is developed to accurately model the cavity. Figure 6 presents a graph illustrating transmissivity (blue line) and reflectivity (red line) as a function of wavelength. The specifications of the cavity are the same as the TMM simulation model. The left axis represents Amplitude (dB), while the right axis denotes the magnitude of the Quality Factor. The Fabry-Perot Bragg Cavity Resonator achieves its highest quality factor of 143,040 at a wavelength of 1293.75 nm. At a wavelength of 1310 nm, the quality factor is measured to be 141,305. Figure 7 depicts the simulation setup in Lumerical INTERCONNECT.

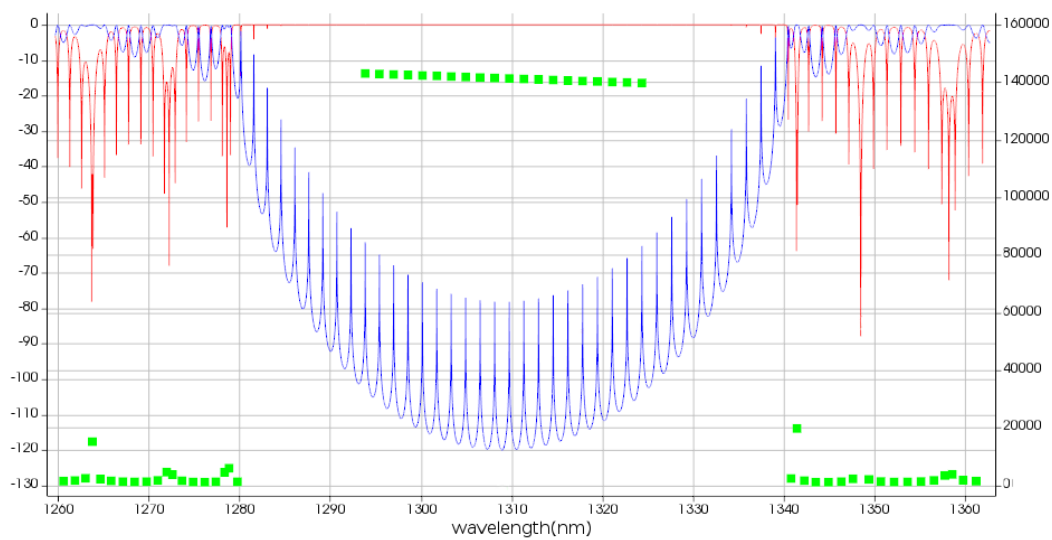


Figure 6: Calculated response of the structure using Lumerical INTERCONNECT

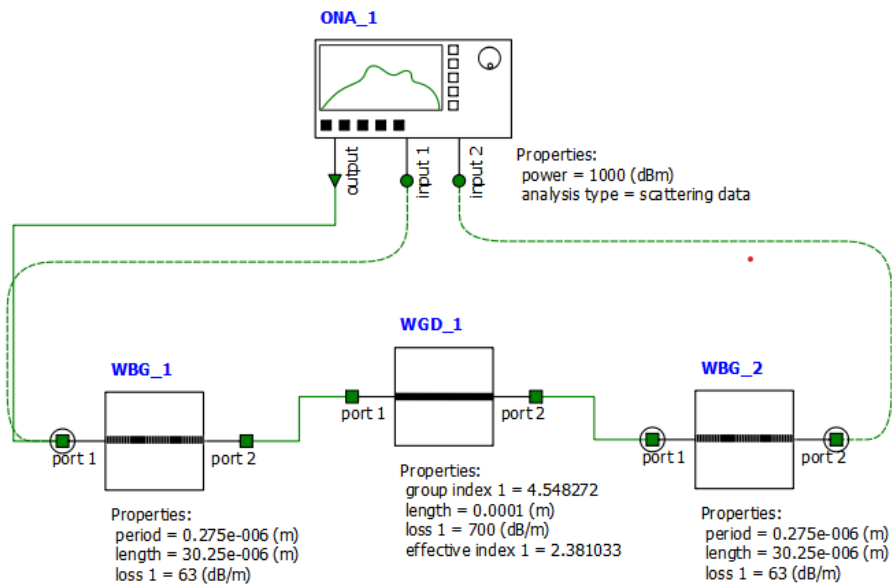


Figure 7: Lumerical INTERCONNECT Simulation Setup

5 Fabrication

KLayout software is employed to design the waveguides and fabricate the Fabry-Perot Cavity resonators using e-beam lithography technology for precise circuit printing. Figure 8 displays an example of the KLayout model. Three grating couplers, spaced 127 μm apart, are necessary to facilitate light input into the circuit and enable output measurements for both transmission and reflection of the cavity. The circuit in Figure 8 also incorporates an adiabatic Y-splitter to introduce light into the cavity and obtain reflectivity data. Bragg gratings are positioned on either side of a waveguide, and multiple instances of the circuit are designed with varying period numbers, cavity length and Bragg period. A summary of all the variations and designs is provided in Tables 4-6 in Section 6.

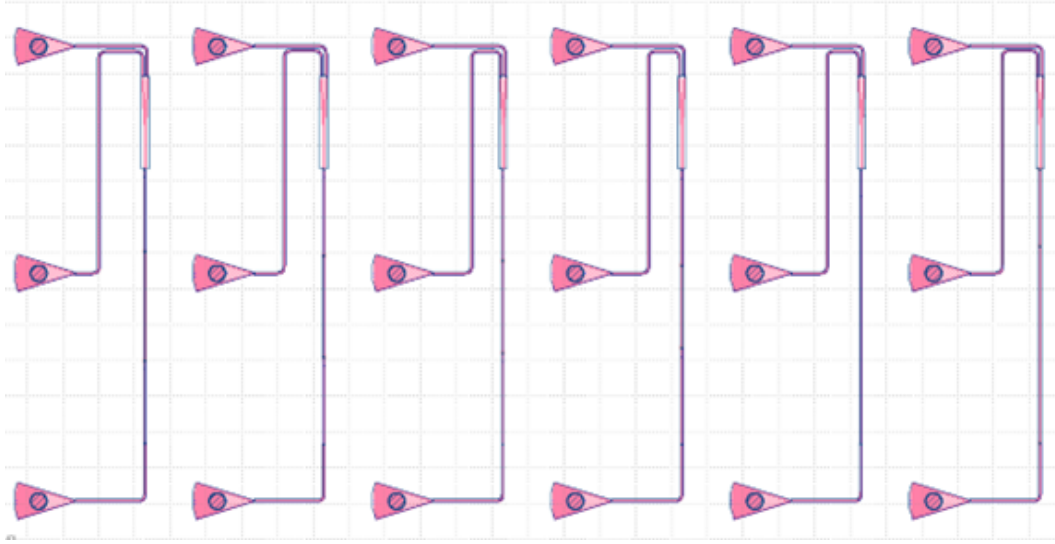


Figure 8: Klayout example: EBeam_{ELEC413}_{AryanC2}

Chips are fabricated by Applied Nanotools utilizing their NanoSOI process, which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator (SOI) wafers with specific dimensions serve as the base material for fabrication. The fabrication process involves pre-dicing the wafer, cleaning it, applying hydrogen silsesquioxane (HSQ) resist, and patterning the photonic devices using an electron beam instrument. Exposure dosage is adjusted to account for proximity effects, and shape writing order is optimized to enhance efficiency and minimize beam drift. Following e-beam exposure and development, the devices are inspected for residues or defects. Subsequently, the chips are mounted on a handle wafer and subjected to an anisotropic ICP-RIE etching process using chlorine. After removing the resist and verifying the patterning and etch quality via scanning electron microscopy (SEM), a 2.2 μm oxide cladding is deposited using a plasma-enhanced chemical vapor deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements are conducted throughout the process to confirm device layer, buffer oxide, and cladding thicknesses prior to delivery.

6 Rationale for Variable Parameters

3 Klayout models have been submitted for fabrication. In two of these layouts the **period number** of a 350 nm wide cavity with a corrugation width of 30 nm and period of 269nm are swept (10 periods in difference). The overall length of the cavity stays the same, therefore as the Bragg number increases the **length of the waveguide cavity** in between decreases (becomes less than 100 μm). In the third one, the period number of a 350 nm wide cavity with a corrugation width of 30 nm and a period of 275nm are swept, the same condition applies to the waveguide as with the other two layouts.

By reiterating the equations above different values of quality factor were obtained. These values are summarized in Table 7.

Table 4: Table of specifications for layout Ebeam_ELEC413_AryanC

Parameter	Value
Fabrication Width	350 nm
Corrugation width - dw	30 nm
Thickness	220 nm
Period	269 nm
Period number	110, 120, 130, 140, 150, 160
Expected quality factor	50,000 – 150,000

Table 5: Table of specifications for layout Ebeam_ELEC413_AryanC2

Parameter	Value
Fabrication Width	350 nm
Corrugation width - dw	30 nm
Thickness	220 nm
Period	269 nm
Period number	170 — 180 — 190 — 200 — all Bragg (no waveguide) — all waveguide(no Bragg)
Expected quality factor	5,000 – 50,000

Table 6: Table of specifications for layout Ebeam_ELEC413_AryanC3

Parameter	Value
Fabrication Width	350 nm
Corrugation width - dw	30 nm
Thickness	220 nm
Period	275 nm
Period number	100, 110, 130, 150, 170, 190
Expected quality factor	> 100,000

Table 7: Quality Factor values from Lumerical INTERCONNECT Simulation

Parameter	Quality Factor at 1310 nm
ARYANC1_1: $\Lambda = 269$ nm, $N = 110$, $L_{\text{cavity}} = 100$ microns	141498
ARYANC1_2: $\Lambda = 269$ nm, $N = 120$, $L_{\text{cavity}} = 94.62$ microns	141246 (159105 at 1308nm)
ARYANC1_3: $\Lambda = 269$ nm, $N = 130$, $L_{\text{cavity}} = 89.24$ microns	150000
ARYANC1_4: $\Lambda = 269$ nm, $N = 140$, $L_{\text{cavity}} = 83.86$ microns	140296
ARYANC1_5: $\Lambda = 269$ nm, $N = 150$, $L_{\text{cavity}} = 78.48$ microns	139363
ARYANC1_6: $\Lambda = 269$ nm, $N = 160$, $L_{\text{cavity}} = 73.10$ microns	54980
ARYANC2_1: $\Lambda = 269$ nm, $N = 170$, $L_{\text{cavity}} = 67.72$ microns	34203
ARYANC2_2: $\Lambda = 269$ nm, $N = 180$, $L_{\text{cavity}} = 62.34$ microns	23782
ARYANC2_3: $\Lambda = 269$ nm, $N = 190$, $L_{\text{cavity}} = 56.96$ microns	9620
ARYANC2_4: $\Lambda = 269$ nm, $N = 200$, $L_{\text{cavity}} = 51.58$ microns	5633
ARYANC2_5: $\Lambda = 269$ nm, $N = 571$, $L_{\text{cavity}} = 0$ microns	n/A
ARYANC2_6: $\Lambda = 269$ nm, $N = 0$, $L_{\text{cavity}} = 432.22$ microns	n/A
ARYANC3_1: $\Lambda = 275$ nm, $N = 100$, $L_{\text{cavity}} = 100$ microns	124306
ARYANC3_2: $\Lambda = 275$ nm, $N = 110$, $L_{\text{cavity}} = 94.5$ microns	141304
ARYANC3_3: $\Lambda = 275$ nm, $N = 130$, $L_{\text{cavity}} = 83.5$ microns	141300
ARYANC3_4: $\Lambda = 275$ nm, $N = 150$, $L_{\text{cavity}} = 72.5$ microns	139589
ARYANC3_5: $\Lambda = 275$ nm, $N = 170$, $L_{\text{cavity}} = 61.5$ microns	20543
ARYANC3_6: $\Lambda = 275$ nm, $N = 190$, $L_{\text{cavity}} = 50.5$ microns	3543

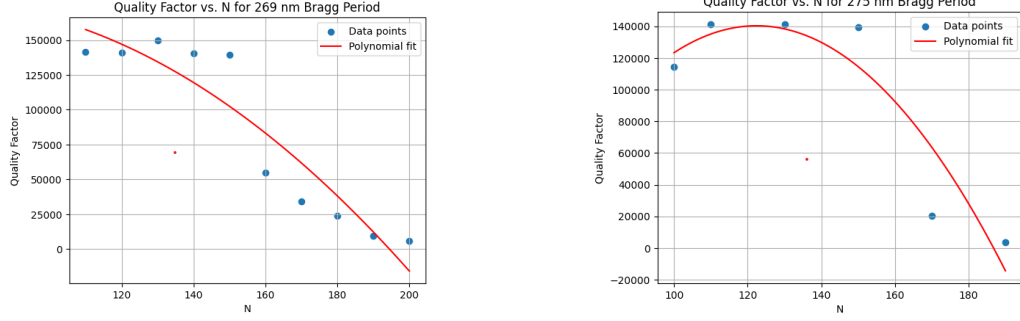


Figure 9: Quality Factor vs Bragg Number Characteristics

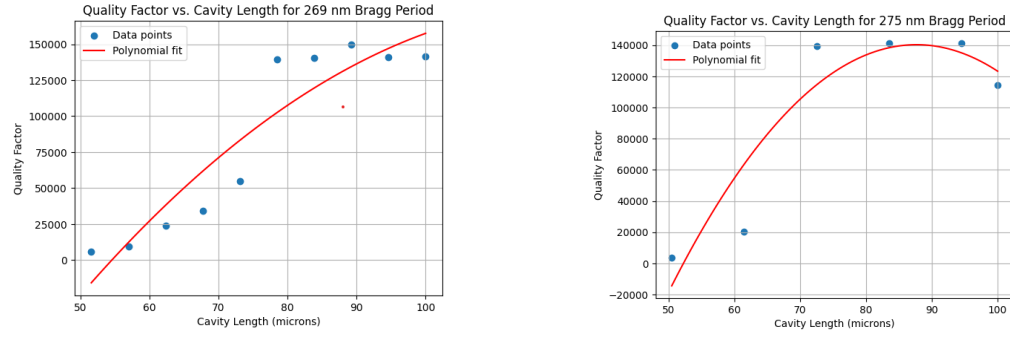


Figure 10: Quality Factor vs Waveguide Cavity Length Characteristics

Upon increasing the Bragg number, the number of periods within the Bragg grating consequently increases, which can lead to enhanced reflection at the Bragg wavelength and improved confinement within the cavity. Nonetheless, the quality factor of a resonator is not solely dependent on these factors; the waveguide cavity length between the two Bragg gratings plays a pivotal role as well. To study the effects of these parameters, the resonator length is maintained constant during the investigation.

As the Bragg number increases, the cavity length between the two Bragg gratings diminishes. A reduced cavity length may result in a higher round-trip loss for the resonant mode due to the augmented coupling between the forward and backward propagating modes. This increase in loss can counterbalance the heightened confinement provided by the additional Bragg periods, thereby leading to a diminished quality factor. Furthermore, as the cavity length decreases, the resonator's free spectral range (FSR) expands, signifying that the spacing between adjacent resonant modes becomes larger. As a result, the resonant modes become more susceptible to minor fluctuations in cavity length, and the quality factor may not exhibit a consistent increase.

The graphs depicting the relationship between N versus quality factor and cavity length versus quality factor for the case with a 275 nm Bragg period do not exhibit a strictly linear increase or decrease. Interestingly, these graphs reveal local minima and maxima within the range of approximately 130 for the N graph and 89.24 microns for the cavity length graph. This intriguing behavior may arise from two potential sources.

Firstly, it is possible that the observed minima and maxima could be an artifact of the polynomial fit employed in the data analysis. In this case, the nonlinearity may not be a genuine feature of the underlying physics but rather an outcome of the chosen fitting method.

Alternatively, the presence of minima and maxima might suggest an optimal combination of cavity length and Bragg number that leads to enhanced performance. In this scenario, there may exist a specific configuration of cavity length and Bragg number that is particularly well-suited for achieving

the desired quality factor. This optimal configuration could result from intricate interactions between various physical properties of the resonator, such as confinement, coupling, and round-trip loss.

The intricate relationship between the Bragg number, cavity length, and quality factor relies on achieving a balance between the enhanced confinement from additional Bragg periods and the increased round-trip loss due to the shortened cavity length. By keeping the resonator length constant, the study focuses on understanding the interplay of these factors and their impact on the overall performance of the resonator.

7 Test Results

A script (Appendix: B1) is used to analyze a set of 12 datasets, visualize the transmission and reflection spectra, and calculate the Q-factor for the reflection spectra.

For each dataset, the code loads the data and extracts the relevant information such as the values for channels 1, 2, and 3, and the wavelength. It then finds the indices of data points within the desired wavelength range of 1.27 to 1.33 micrometers.

The Q-factor is calculated by obtaining the wavelength and reflection data as inputs. Then all the local maxima in the reflection spectrum is found using the findpeaks function. For each peak, the function calculates the Q-factor as follows:

Identify the half maximum value for the current peak (half of the peak value). Find the indices where the reflection is equal to the half maximum value, both to the left and right of the peak. These indices correspond to the points where the reflection spectrum crosses the half maximum value. Calculate the full width at half maximum (FWHM) by subtracting the wavelengths at the two closest indices found in step 2. Determine the center wavelength as the average of the two closest wavelengths found in step 2. Calculate the Q-factor as the ratio of the center wavelength to the FWHM and append it to the QFactors array. After calculating the Q-factors for all peaks in the reflection spectrum, the script finds the maximum Q-factor and plots the transmission and reflection spectra in a subplot. It also adds the maximum Q-factor as text beneath the graph.

In summary, this script provides a comprehensive analysis of the datasets, displaying the calibrated transmission and reflection spectra and calculating the Q-factors for the reflection spectra. The Q-factor is a measure of the quality of a resonator and is calculated as the ratio of the center wavelength to the full width at half maximum of the resonance peak.

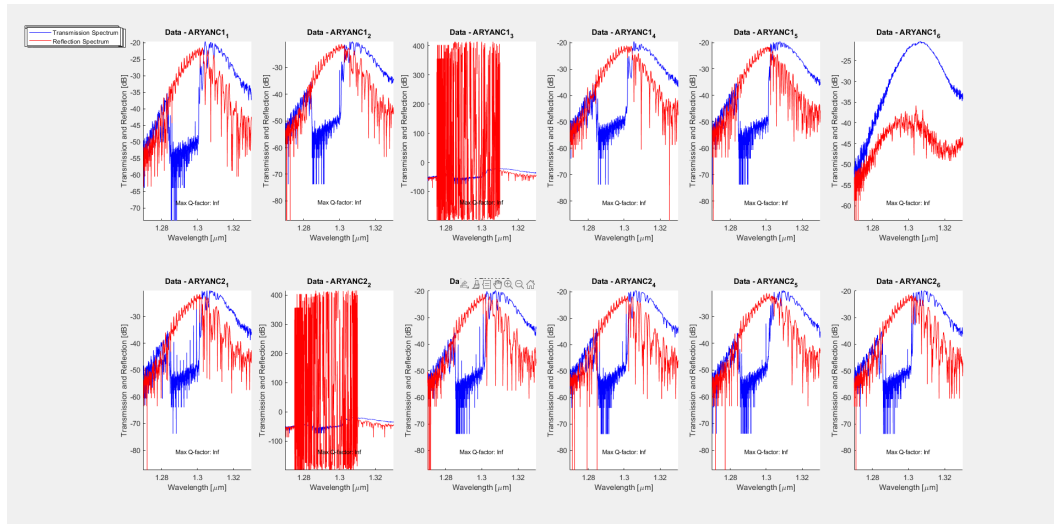


Figure 11: EBeamELEC413AryanC1 and EBeamELEC413AryanC2 Results Without Calibration

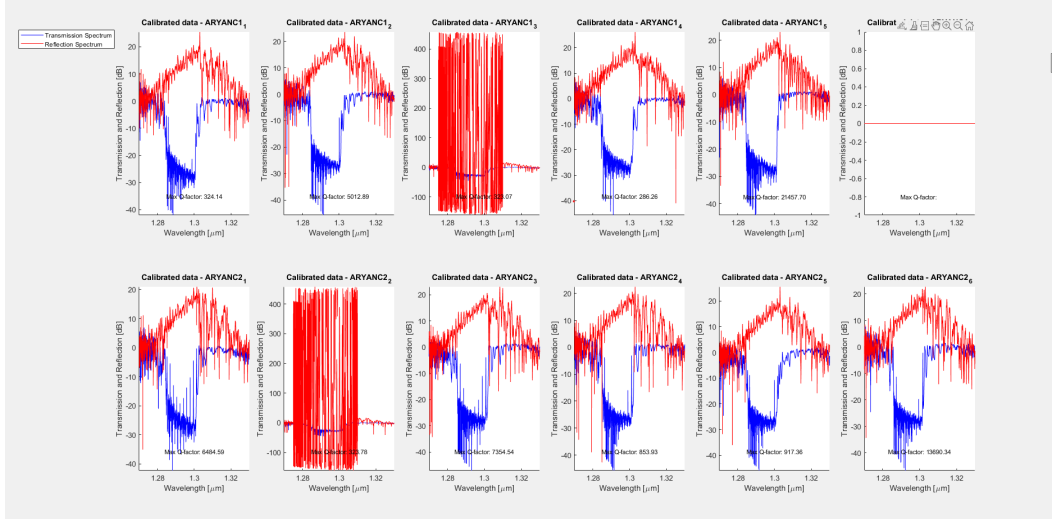


Figure 12: EBeamELEC413AryanC1 and EBeamELEC413AryanC2 Results With Calibration

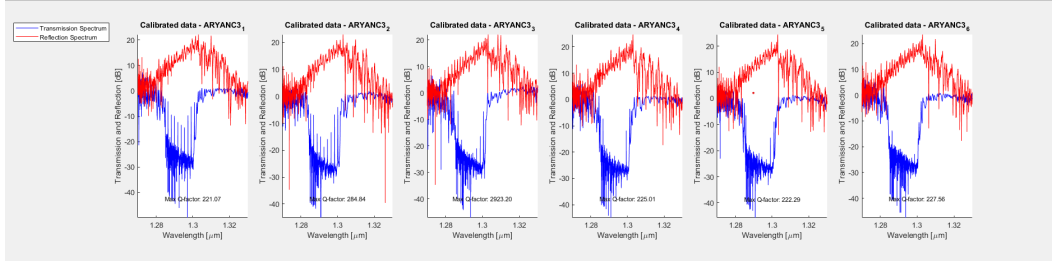


Figure 13: EBeamELEC413AryanC3 Results With Calibration

8 Conclusion

In conclusion, our experimental investigation revealed that the interplay between the Bragg number, cavity length, and quality factor is crucial for optimizing the performance of resonators. Our results demonstrated that increasing the Bragg number leads to enhanced reflection at the Bragg wavelength and improved confinement within the cavity. However, the waveguide cavity length between the two Bragg gratings also plays a pivotal role in determining the quality factor of the resonator. Achieving a balance between enhanced confinement and increased round-trip loss due to shortened cavity length is essential for optimal performance. The highest quality factor achieved is for the design ARYANC1_5 (21457).

Although there were discrepancies between the analytical and experimental quality factors, our predictions regarding the effects of Bragg number and cavity length on the quality factor remained valid. The presence of local minima and maxima in the graphs suggests an optimal combination of cavity length and Bragg number that leads to enhanced performance. This optimal configuration could result from intricate interactions between various physical properties of the resonator, such as confinement, coupling, and round-trip loss.

Furthermore, we discussed the potential impact of the fabrication process on the resonator's performance. The presence of an oxide layer can cause a shift in the central frequency, which affects the resonance behavior of the device. It is important to consider such effects while designing and optimizing the performance of resonators.

We also highlighted the need for calibration to ensure accurate and reliable experimental results. Although Design ARYANC2_6 was originally intended for calibration, we found that Design ARYANC1_6 exhibited smoother results, so we used that instead. However, some designs showed

inconclusive data, such as ARYANC2_2 and ARYANC1_3, which could be due to various factors such as fabrication variations, measurement errors, or intrinsic device characteristics. Further investigations and optimizations are required to understand and improve the performance of such designs.

In summary, our study provides valuable insights into the interplay between the Bragg number, cavity length, and quality factor in resonators. Our findings can be useful for future research and practical applications of resonators in various fields. It is crucial to understand the intricate relationship between these parameters to optimize the performance of resonators. The knowledge gained from our study can inspire further investigations and advancements in the field of resonators.

A 1

```
# Get the data from the MODE frequency sweep – Provided by Lukas Chrostowski
as Part of ELEC 413 UBC Course
lambda=c/getdata("FDE::data::frequencysweep","f")*1e6; # lambda in [microns]

neff=abs(getdata("FDE::data::frequencysweep","neff"));
lambda0 = 1.55; # central wavelength

# Polynomial fitting.
x_fit = lambda-lambda0; # X vector
y_fit = neff;           # Y vector
n_fit = 2;              # order of the polynomial

# Next 3 lines do polyfit. Inputs: x_fit, y_fit, n_fit. Outputs: a_fit.
# Based on a linear algebra approach
X_fit = matrix(length(x_fit),n_fit+1);
for(i_fit=0:n_fit){ X_fit(1:length(x_fit),i_fit+1) = x_fit^i_fit; }
a_fit = mult(mult(inv(mult(transpose(X_fit),X_fit)),transpose(X_fit)),y_fit);

# Display the fit polynomial coefficients
?a_fit;

# This is the polynomial equation:
Y_fit_result = a_fit(1) + a_fit(2)*(lambda-lambda0) + a_fit(3)*(lambda-lambda0)^2;

plot(lambda, neff, Y_fit_result); # plot the result
```

B 1

```
% Preallocate an array to store the file names
fileNames = cell(1, 12);

% Generate the file names for ARYANC1_X and ARYANC2_X
for i = 1:6
    fileNames{i} = sprintf('ARYANC1_%d', i);
    fileNames{i+6} = sprintf('ARYANC2_%d', i);
end
% Load the calibration data from ARYANC2_6.mat
calibrationData = load('ARYANC1_6.mat');
calibrationCH2 = calibrationData(1,1).testResult.rows.channel_2;
calibrationCH3 = calibrationData(1,1).testResult.rows.channel_3;

% Create a new figure
figure;

% Loop through each file and load the data
for i = 1:12
    % Load the data
    Data = load(fileNames{i});

    % Extract the relevant data
    CH1 = Data(1,1).testResult.rows.channel_1;
    CH2 = Data(1,1).testResult.rows.channel_2 - calibrationCH2;
    CH3 = Data(1,1).testResult.rows.channel_3 - calibrationCH3;
    lambda = Data(1,1).testResult.header.wavelength/1e9;
```

```

% Find indices of data points within the desired wavelength range
idx = find(lambda*1e6 >= 1.27 & lambda*1e6 <= 1.33);

% Calculate Q-factors for the r spectrum
QFactors = calculate_QFactor(lambda(idx), CH3(idx));

% Find the maximum Q-factor
maxQFactor = max(QFactors);

% Create a subplot for each dataset in a 2x6 grid
subplot(2, 6, i);
title(sprintf('Calibrated data - %s', fileNames{i}));
hold on;
plot(lambda(idx)*1e6, CH2(idx), 'b');
plot(lambda(idx)*1e6, CH3(idx), 'r');
hold off;
xlabel('Wavelength [\mu m]');
ylabel('Transmission and Reflection [dB]');
legend('Transmission Spectrum', 'Reflection Spectrum');
axis tight;
xlim([1.27, 1.33]);

% Add the maximum Q-factor text beneath the graph
xRange = xlim;
yRange = ylim;
text(xRange(1) + 0.5 * (xRange(2) - xRange(1)),
yRange(1) + 0.1 * (yRange(2) - yRange(1)),
sprintf('Max Q-factor: %.2f', maxQFactor),
'HorizontalAlignment', 'center', 'FontSize', 8);
end

function QFactors = calculate_QFactor(lambda, transmission)
% Find all local minima in the transmission spectrum
[~, minimaIdx] = findpeaks(-transmission);

% Initialize the QFactors array
QFactors = [];

% Loop through all minima to calculate the Q-factor for each peak
for i = 1:length(minimaIdx)
    % Find the half maximum value for the current peak
    halfMaxVal = transmission(minimaIdx(i)) / 2;
    % Find indices where transmission is equal to the half maximum value
    % around the current peak
    halfMaxIdx = find(transmission >= halfMaxVal, 1, 'first');
    halfMaxIdx(end+1) =
        find(transmission(halfMaxIdx:end)
        <= halfMaxVal, 1, 'last') + halfMaxIdx - 1;
    % Find the FWHM by subtracting the wavelengths at the two closest indices
    FWHM = lambda(halfMaxIdx(end)) - lambda(halfMaxIdx(1));
    % Calculate the center wavelength as the average of the
    % two closest wavelengths
    centerWavelength = (lambda(halfMaxIdx(end)) + lambda(halfMaxIdx(1))) / 2;
    % Calculate the Q-factor and append it to the QFactors array
    QFactors(end+1) = centerWavelength / FWHM;
end
end

```

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